GEOMETRIC REGULARITY DESCENDS ALONG A FLAT MORPHISM OF LOCALLY NOETHERIAN SCHEMES OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter text immediately before the lemma notes that it is a geometric variant of an algebra lemma; that remark lies outside the counted span and enters only as context.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-flat-under-geometrically-regular is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Regularity descends along flat local homomorphisms of Noetherian local rings. Algebra, Lemma algebra-lemma-flat-under-regular, invoked at proof lines 2537-2540. Supplies the descent of regularity on local rings.
- The pointwise criterion for geometric regularity. Same chapter, Lemma lemma-geometrically-regular, invoked at proof lines 2525-2527. Supplies the passage from regularity after all finite purely inseparable base changes to geometric regularity.
- Flatness is preserved by base change. Morphisms, Lemma morphisms-lemma-base-change-module-flat, invoked at proof lines 2528-2530. Supplies that the base changed morphism is still flat at the lifted point.

Source reference. Original lemma 00OF: sources/stacks-chow/tags.txt, source lines 893–893. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038V: sources/stacks-varieties/pinned-original-source.tex, source lines 2424–2424. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01U8: sources/stacks-chow/tags.txt, source lines 2401–2401. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let $f : X \to Y$ be a morphism of locally Noetherian schemes over k. Let $x \in X$ be a point and set $y = f(x)$. If X is geometrically regular at x and f is flat at x then Y is geometrically regular at y. In particular, if X is geometrically regular over k and f is flat and surjective, then Y is geometrically regular over k.*

Proof. Let k' be finite purely inseparable extension of k. Let $f' : X_{k'} \to Y_{k'}$ be the base change of f. Let $x' \in X_{k'}$ be the unique point lying over x. If we show that $Y_{k'}$ is regular at $y' = f'(x')$, then Y is geometrically regular over k at y', see Lemma 038V. By Morphisms, Lemma 01U8 the morphism $X_{k'} \to Y_{k'}$ is flat at x'. Hence the ring map

$$\mathcal{O}_{Y_{k'},y'} \longrightarrow \mathcal{O}_{X_{k'},x'}$$

is a flat local homomorphism of local Noetherian rings with right hand side regular by assumption. Hence the left hand side is a regular local ring by Algebra, Lemma 00OF. □